MATLAB 绘图

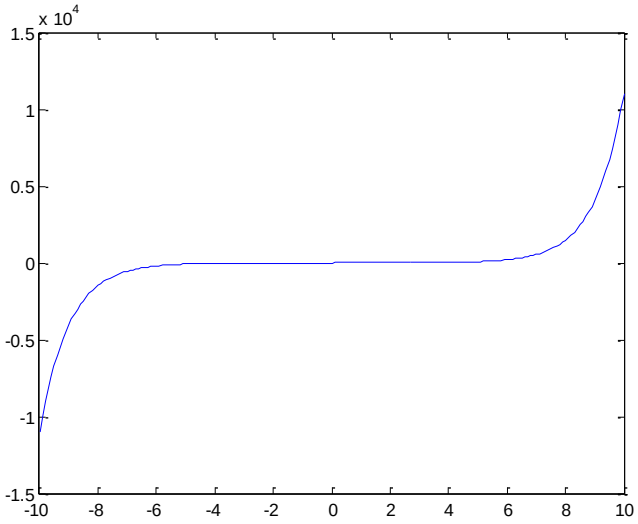
强调三个问题

1. 区分三维曲面和曲线作图
2. 要区分隐函数作图
3. 符号函数作图

第三次课的作业题

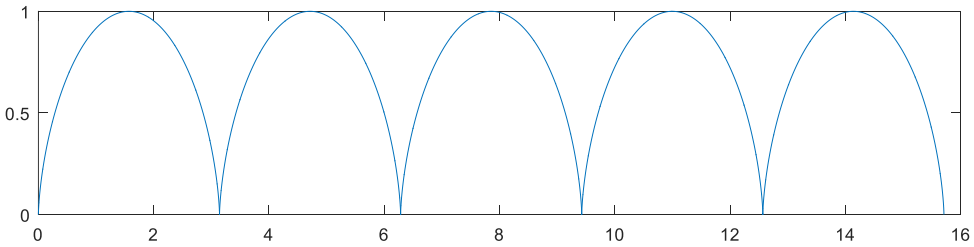
5 (1) $y=\frac{a}{2}(e^{\frac{x}{a}}-e^{-\frac{x}{a}})$；

```
x=-10:0.1:10;
a=1;
y=1/2*(exp(x/a)-exp(-x/a));
plot(x,y)
```



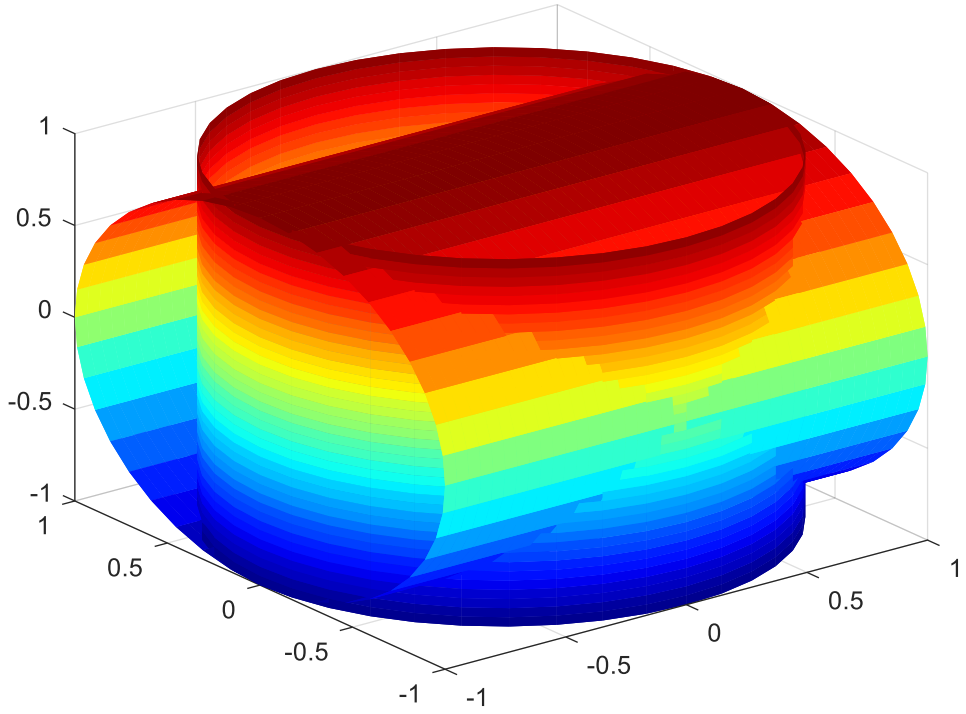
(2) $\begin{cases} x=a(t-\sin t) \\ y=a(1-\cos t) \end{cases}$；

```
a=0.5;
t=0:0.01:10*pi;
x=a*(t-sin(t));
y=a*(1-cos(t));
plot(x,y)
```



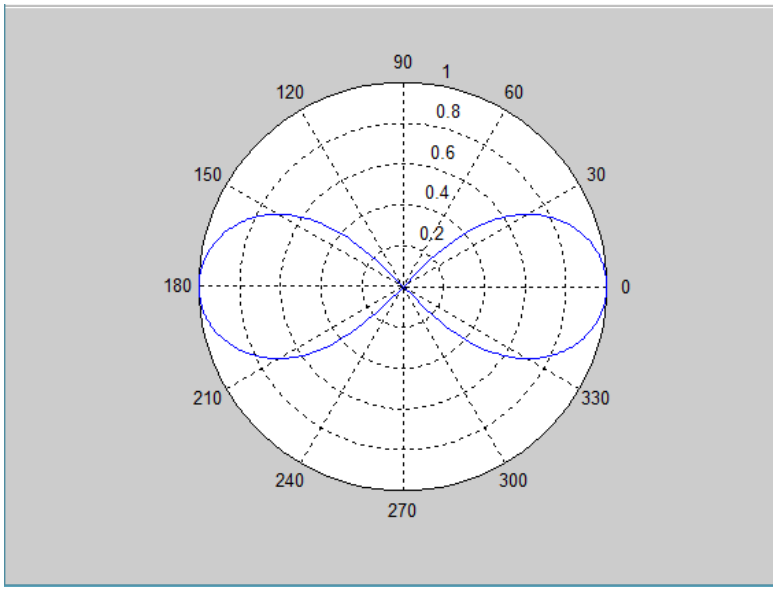
(3) $\begin{cases} x^2 + y^2 = 1 \\ y^2 + z^2 = 1 \end{cases}$

```
clc,clear
t=0:pi/20:2*pi;
x=sin(t);
y=cos(t);
z=linspace(-4,4,length(t));
X=meshgrid(x);Y=meshgrid(y);Z=[meshgrid(z)]';
surf(X,Y,Z)
hold on
surf(Z,X,Y)
```



(4) $r^2 = a^2 \cos 2\theta$;

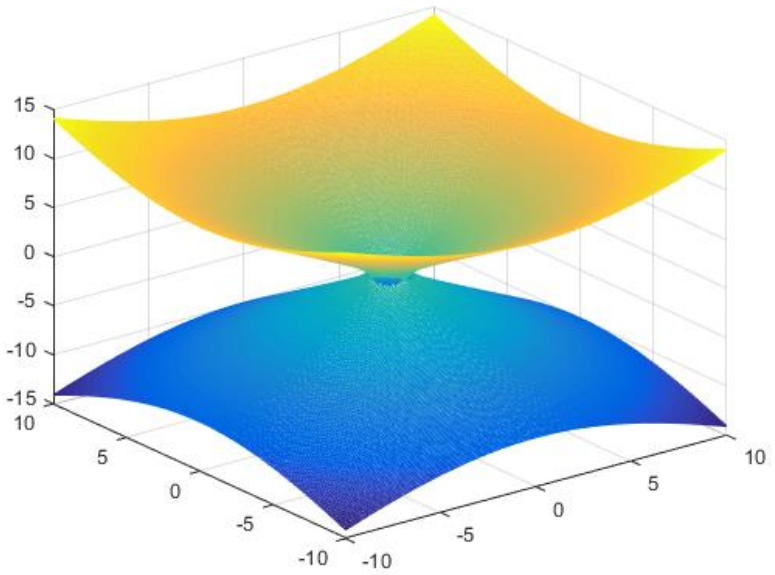
```
t=0:0.01:2*pi;
a=1;
r=sqrt(a^2*cos(2*t))
r(cos(2*t)<0)=NaN;
polar(t,r);
```



(5) $x^2 + y^2 - z^2 = 1$;

```
x=-10:0.1:10;
y=x;
[X,Y]=meshgrid(x,y);
Z=sqrt((X.^2+Y.^2)-1);
Z((X.^2+Y.^2)-1<0)=NaN;
mesh(X,Y,Z)
hold on
Z2=-sqrt((X.^2+Y.^2)-1);
Z2((X.^2+Y.^2)-1<0)=NaN;
mesh(X,Y,Z2)
```

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（6） $z = y^2 + xy$

```
[x,y]=meshgrid(-10:0.1:10);
z=y.^2+x.*y;
mesh(x,y,z)
```

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